Food Security Status Across the Rural-Urban Continuum Before and During the COVID-19 Pandemic

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Abstract

Background: Food security, defined as consistent access to sufficient food to support an active life, is a crucial social determinant of health. A key dimension affecting food security is position along the rural-urban continuum, as there are important socio-economic and environmental differences between communities related to urbanicity or rurality that impact food access. The COVID-19 pandemic created social and economic shocks that altered financial and food security, which may have had differential effects by rurality and urbanicity. However, there has been limited research on how food security differs across the shades of the rural-urban community spectrum, as most often researchers have characterized communities as either urban or rural.

Methods: In this study, which linked restricted use Current Population Survey Food Security Supplement data to census-tract level United States Department of Agriculture Rural-Urban Commuting Area codes, we estimated the prevalence of household food security across temporal (2015-2019 versus 2020-2021) and socio-spatial (urban, large rural city/town, small rural town, or isolated rural town/area) dimensions in order to characterize variations before and during the COVID-19 pandemic by urbanicity/rurality. We report prevalences as point estimates with 95% confidence intervals.

Results: The prevalence of food security was 87.7% (87.5-88.0%) in 2015-2019 and 88.8% (88.4-89.3%) in 2020-2021 for urban areas, 85.5% (84.7-86.2%) in 2015-2019 and 87.1% (85.7-88.3%) in 2020-2021 for large rural towns/cities, 82.8% (81.5-84.1%) in 2015-2019 and 87.3% (85.7-89.2%) in 2020-2021 for small rural towns, and 87.6% (86.3-88.8%) in 2015-2019 and 90.9% (88.7-92.7%) in 2020-2021 for isolated rural towns/areas.

Conclusion: These findings show that rural communities experiences of food security vary and aggregating households in these environments may mask areas of concern and concentrated need.

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Introduction

Food security is a social determinate of health characterized by adequate access to sufficient amounts of food to support an active lifestyle (Bickel et al, 2000). Experiencing food insecurity is associated with many negative health consequences (Gundersen & Ziliak, 2015; Seligman et al 2010) and increased costs for communities (Berkowitz et al, 2019; Dean et al, 2020). Therefore, ensuring adequate access to quality food must be a public health priority. Many factors contribute to household food security status, but the primary driver of food insecurity in the US, a country that relies on the globalized and market-based food system, is the failure to ensure adequate economic resources for all households (Bickel et al, 2000; Drewnowski, 2022). In the US, food assistance, particularly the Supplemental Nutrition Assistance Program (SNAP) but also other government food programs such as the Women, Infants, and Children Program and National School Lunch Program, are effective at reducing food insecurity (Huang & Barnridge, 2016; Gundersen, Kreider, & Pepper, 2017; Kreider, Pepper, & Roy, 2016; Nord et al, 2009), but these resources have not fully eliminated it (Coleman-Jensen et al, 2021). This may be due to inadequate benefits that do not fully cover what recipients need to be food secure, underuse by eligible households, and ineligibility of many households with unmet food needs. Additionally, emergency food providers in the United States provide support to many households with inadequate food access, but there is little evidence on the impact of emergency food on food security rates at the population level (Gundersen & Ziliak, 2018).

Food security rates vary along the rural-urban continuum. In 2020 the overall food security rate in the U.S. was 89.5%, but rates were 87.3% in urban cores of metropolitan areas and 88.4% in non-metropolitan areas, including rural areas, compared to 91.2% for non-core metropolitan, exurban, and suburban areas (Coleman-Jensen, et al., 2021). This is partly due to economic factors in these communities, such as persistent poverty in urban centers and rural communities (Nolan et al 2017) and the migration of working-age adults from rural communities to more urbanized areas in pursuit of economic opportunities (Johnson & Lichter, 2019). Healthy food environments consist of the collection of norms, economic conditions, settings and organizations, and policies determined by sociocultural and

political forces that enable the capability to consume a health promoting diet (Swinburn, Egger, & Raza et al, 1999). Differences in the quantity and types of food retailers and other food-related resources also likely contributes to these observed differences, where core urban and rural communities often lack healthy food environments (Cakir et al, 2020; Gantner et al, 2011; Larson et al, 2009).

Food security rates also often coincide with major economic trends, e.g., decreasing substantially following the 2007-08 financial crisis, reaching a nadir in 2011, and steadily increasing between 2012 and 2019 (Coleman-Jensen, et al., 2021). In early 2020 when the COVID-19 pandemic emerged, many expected a steep rise in food insecurity due to the system-wide economic shock, loss of employment, supply chain disruptions, and general market uncertainty. Despite evidence that households experienced initial increased food insecurity due to economic disruptions caused by the COVID-19 pandemic (Gundersen et al, 2021; Niles et al, 2020; Willis et al 2022), 12-month food security rates were stable between 2019 and 2020 at approximately 89.5% of U.S. households (Coleman-Jensen, et al., 2021). However, these overall rates obscure some disparities; the prevalence of households experiencing food security decreased among lower income, Black, non-Hispanic and Hispanic, and single parent households, indicating inequitable impact of the pandemic by social and economic position (Coleman-Jensen, et al., 2021). These trends may be explained by the substantial levels of support that were implemented in response to the pandemic, but with a possible unevenness in the distribution of these resources (Kinsey et al 2020).

We aimed to better understand relationships between food security and the rural-urban continuum before and during the COVID-19 pandemic. While previous analyses have examined rural/urban differences in food security at the county level (Gundersen, Dewey, Hake, Engelhard, & Crumbaugh, 2017), there are important sub-county level differences in communities that affect food access and food security status security. Rural-Urban Commuting Area (RUCA) codes offer the opportunity for examining food insecurity at the census-tract level while also considering commuting flows in urban/rural classifications. Accounting for commuting flows, rather than just physical location of a household, captures the environments that community members experience in their day-to-day lives.

We extend previous work on geographic patterns of food security by using a sub-county level measure and introduce the context of the COVID-19 pandemic into our understanding of this phenomenon. We answer the following two research questions: 1) What is the prevalence of households who experienced food security at each level of the rural-urban continuum defined by RUCA codes before and during the COVID-19 pandemic, both overall and for low-income households? and 2) What are the relationships between demographic factors, time period, and rural/urban status with experiencing food security overall and for low-income households?

Data and Methodology

Data

The Current Population Survey (CPS) Food Security Supplement (FSS) is a collaboration between the United States Census Bureau, Bureau of Labor Statistics, and United States Department of Agriculture that provides annual data related to unmet food needs and use of food assistance resources among US households. Each December, approximately 30,000 households participate. We linked the restricted use CPS FSS to publicly available United States Department of Agriculture 2010 Rural-Urban Commuting Area (RUCA) Codes (available from https://www.ers.usda.gov/data-products/rural-urban-commuting-area-codes/). These data contain both primary and secondary codes that classify census tracts based on their population density, urbanicity, and commuting flows. Households that completed the CPS FSS were assigned primary and secondary RUCA codes by their census tract identifier.

Design

We used a repeated cross-sectional design. All analyses were adjusted for survey design and non-response using household-level weights provided in the CPS FSS.

Sample

Our analytic sample consisted of households who completed the FFS between 2015 and 2021. Households that completed the FSS in two consecutive years had their second responses excluded from the analysis. There were approximately 135,000 households in our analytic sample. We also conducted sub-analyses with households earning under 185% of the Federal Poverty Threshold, estimated using the

midpoint of each household's income category and the household size. There were approximately 38,500 households in the low-income sub-sample.

Measures

Food security status was measured using the 18-item United States Department of Agriculture Household Food Security Survey Module (HFSSM). Households were classified as either food secure or food insecure in the CPS FSS based on their responses using USDA scoring methods (Bickel et al, 2000). Time period was defined as "Pre-COVID" for 2015-2019 and "During COVID" for 2020 and 2021. The RUCA codes consist of primary and secondary codes that classify census tracts on the rural-urban continuum based on population size, urbanization, and daily commuting (USDA-ERS, n.d.). The primary RUCA codes include 10 levels and only consider the primary commuting flow, or the single largest commuting destination. Secondary RUCA codes also consider secondary commuting flows, or the second largest commuting destination. For our analysis, we used secondary RUCA codes and classified households as urban, large rural city/town, small rural town, and isolated rural town/area. Table 1 shows the classification scheme based on the secondary RUCA codes. We also examined the data using a primary RUCA coding scheme (coding 1-3 as metropolitan, 4-6 as micropolitan, 7-9 as small town, and 10 as rural), but did not present the result here as they provided similar estimates.

In adjusted analyses, we controlled for the following demographic characteristics: household size (continuous), age of household respondent (34 and under, 35-59, 60 and older), sex of household respondent (Male or Female), racial self-classification of household respondent (White; Black; Asian, American Indian/Alaskan Native, Native Hawaiian\Pacific Islander; or Multiracial), Hispanic ethnicity of household respondent (Hispanic or non-Hispanic), household income (less than $5,000, $5,000-$7,499, $7,500-$9,999, $10,000-$12,499, $12,500-$14,999, $15,000-$19,999, $20,000-$24,999, $25,000-$29,999, $30,000-$39,999, $40,000-$49,999, $50,000 or more), and employment status of household respondent (employed, unemployed, or not in labor force). We include the sex, racial self-classification, and Hispanic ethnicity of the household respondent variables as proxies for exposure to systemic and

structural barriers that constrain food and economic security and access to resources (Boyd, Lindo, Weeks, McLemore, 2020).

Analysis

RQ1.

Using the combination of data sources described previously, we estimated the prevalence of food secure households within each category of our measure of the rural-urban continuum and examined overall unadjusted pandemic-related changes over time according to the following equation (EQ1) for all households and households with incomes less than approximately 185% of the Federal Poverty Threshold:

$$Weightedprevalence \in timeperiod\ j \wedge urban/rural\ codek = \frac{\sum_i y_{ijk} * w_{ijk}}{\sum_i w_{ijk}} \quad \text{(EQ1)}$$

where y_{ijk} is a binary measure of food insecurity for household *i*, in time period *j*, and in urban/rural code *k* and where w_{ijk} is the Food Security Supplement household weight for household *i*, in time period *j*, and in urban/rural code *k*.

RQ2.

Next, to examine the relationship between demographic factors, time period, and urban/rural status, we estimated the following logistic regressions for household food security status (EQ2) applying Food Security Supplement household weights to account for survey design and non-response for all households and households with incomes less than approximately 185% of the Federal Poverty Threshold.

$$E(y\ \ ) = \alpha + \beta_1 t_j + \beta_2 r_k + \beta_3 (t_j * r_k) + \beta_4 State + \beta_5 X_i + \epsilon \quad \text{(EQ2)}$$

where y_{it} is the log odds of a household experiencing food security for household *i* in time *t*, t_j is the time period (2015-2019 or 2020-2021), r_k is the urban/rural code (urban, large rural city/town, small rural town, and isolated rural town/area), State is a fixed effect for state of residence, and $\boldsymbol{X}_i$ represents a vector of demographic characteristics described above for household *i*. The two-way interaction term between t_j and r_k is included to evaluate whether the outcomes varied by time period and urban/rural code.

To examine associations between each outcome and variables of interest accounting for differences in demographics, we calculated the predicted probability of experiencing food security and 95% confidence intervals using the parameter estimates from the logistic regression models.

Results

Sample Characteristics

Table 2 shows the sample demographics. Most households were classified as urban. Overall, approximately nine in ten households experienced food security, while seven in ten low-income households experienced food security.

Food security prevalence estimates

Table 3 displays the prevalences of household experiencing food security in the two timeframes. Overall, unadjusted food security prevalences in 2015-2019 were highest in urban areas (87.7% (95% CI: 87.5-88.0%)) and isolated rural towns/areas (87.6% (95% CI: 86.3-88.8%)). They were slightly lower in large rural towns/cities, 85.5% (95% CI: 84.7-86.2%), and small rural towns, 82.8% (95% CI: 81.5-84.1%). In 2020 and 2021, food security increased across all levels of urbanicity/rurality, with the largest increase for small rural towns to 87.3% (95% CI: 85.7-89.2%). When just considering low-income households, unadjusted food security prevalences were lower and between 75.2% (95% CI: 72.5-77.7) for isolated rural towns/areas and 70.2% (95% CI: 67.8-72.7%) for small rural towns in 2015-2019. Similar to the overall population, the prevalence of food secure low-income households increased in 2020 and 2021 and especially for isolated rural towns/areas, to 81.2% (95% CI: 76.1-85.3%) and small rural towns, to 76.9% (95% CI: 72.6-80.8%. After adjustment for covariates, food security prevalences overall and for low-income households were similar across urban areas, large rural city/town, small rural town, and isolated rural town/area and trends between time periods were similar to the unadjusted estimates.

Conclusions and Future Work

This analysis provided estimates of food security prevalences across the urban-rural continuum before and during the COVID-19 pandemic. We observed either similar or higher prevalence of food security during pandemic (2020 and 2021) time period compared to pre-pandemic (2015-2019). While

there were immediate concerns that the economic disruptions caused by the pandemic were going to severely increase the number of households that experienced food insecurity (Gundersen et al, 2021; Niles et al, 2020; Willis et al 2022), the first two years of the COVID-19 pandemic also included substantial economic supports for households and community-based social service providers, which may have blunted this impact at the population level.

Our findings also reveal important information on the distribution of food security and inadequate economic resources across nuanced levels of the rural-urban continuum. According to annual estimates, food security prevalences for households outside of metropolitan areas were 84.0% in 2019 and 88.4% in 2020 (Coleman-Jensen et al., 2020, 2021). In our analysis, the prevalence of food secure households sometimes differed between large rural towns and cities, small rural towns, and isolated rural towns/areas, showing variation in the distribution of inadequate food access in non-metropolitan areas. Particularly small rural towns before the pandemic appear to be sites of concentrated socioeconomic deprivation, as indicated by a lower pre-pandemic food security prevalence compared to the other rural classifications in unadjusted estimates for all households. These results also suggest that households in small rural towns may have benefited from pandemic-related supports to a higher degree compared to households in other classifications. The reasons for this should be investigated. Addressing food security as a socially produced condition will require addressing systemic factors that disadvantage certain groups, e.g., differential exposure to systems of oppression based on socioeconomic status, race, ethnicity, gender, or age. How rurality/urbanicity interacts with exposure to systemic oppression due to other factors is another area for future investigation.

A strength of this analysis was using RUCA codes to construct more specific geographic variables to classify households' rural status. Overall, there is a lack of data on food access and security in communities at different levels of rurality. These measures allow us to better differentiate between types of rural communities and offer more precision when classifying households as they are at the census tract rather than county level. Additionally, accounting for commuting flows may more accurately reflect the food environments households are able to access (Kerstens et al, 2010). This may be especially important

for individuals living in rural areas where households may access food retailers where they work or regularly travel to rather than where their residence is located. That being said, aggregating households also has limitations for understanding households' experiences of food security. For example, our urban classification included many suburban communities adjacent to major metropolitan areas, which has the effect of obscuring underserved nutritional environments and persistent poverty that are often present in core metropolitan areas (Coleman-Jensen et al., 2020, 2021). Further work that examines food security at the intersection of urbanicity/rurality and socioeconomic status (e.g., classifying households by both RUCA code and a community-level socioeconomic indicators such as median income within each census tract) is warranted.

A more nuanced awareness of the inequitable distribution of adequate food access across differing communities is critical for monitoring progress towards eliminating food insecurity in the United States. Relying on a metropolitan/non-metropolitan dichotomization or county-level estimates does not capture important differences between communities. Understanding where households are most at risk for experiencing food insecurity will have important implications for developing, implementing, and evaluating policy and community-based interventions seeking to enhance households' capabilities to acquire enough food. This is especially true in times of crisis and when large scale changes occur in the nutrition safety net, both of which occurred during the COVID-19 pandemic. Inadequate food access is present in communities across the rural-urban continuum in the United States, but a better understanding the inequitable distribution of households experiencing food security will enable a stronger and targeted response to address this issue.

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Table 1. Rural classification based on Secondary Rural-Urban Area Commuting Codes

Rural Classification	Description	Rural-Urban Area Commuting Codes
Urban	Households living in metropolitan area or households living in micropolitan areas, small rural towns, or isolated rural towns/areas with substantial commuting flows to metropolitan areas	1.0, 1.1, 2.0, 2.1, 3.0, 4.1, 5.1, 7.1, 8.1, 10.1
Large rural city/town	Households living in micropolitan area or households in small rural towns or isolated rural towns/areas with substantial commuting flows to micropolitan areas	4.0, 5.0, 6.0, 7.2, 8.2, 10.2
Small rural city/town	Households living in small rural towns or households in isolated rural towns/areas with substantial commuting flows to small rural towns	7.0, 8.0, 9.0, 10.3
Isolated rural town/area	Households living in isolated rural towns/areas without substantial commuting flows to metropolitan areas, micropolitan areas, or small rural towns.	10.0

Table 2. Descriptive statistics for households overall and for households with incomes less that approximately 185% of the Federal Poverty Threshold from pooled data from the Current Population Survey Food Security Supplement from 2015 to 2021

Characteristic	All households	Low-income households
Rural status, %		
Urban	84.7	80.5
Large rural city/town	8.8	10.8
Small rural city/town	3.7	5.2
Isolated rural town/area	2.8	3.5
Time period, %		
2015-2019	71.5	72.8
2020 & 2021	28.5	27.2
SNAP, %		
Participant	9.4	27.4
Non-participant	90.6	72.6
Emergency food, %		
Received	5.6	14.5
Did not receive	94.4	85.5
Age category, %		
Under 35	23.3	28.5
35-59	43.1	37.3
60 and older	33.6	34.2
Gender, %		
Male	49.2	41.4
Female	50.8	58.6
Race, %		
White	67.6	61.2
Black	12.0	17.7
Asian, American Indian/Alaska Native, Native Hawaiian/Pacific Islander	5.9	6.0
Multiracial	14.5	155.1
Hispanic ethnicity, %		
Yes	14.2	22.5
No	85.8	77.5
Yearly income, %		
Less than $5000	3.0	10.3
$5000-$7499	1.5	4.9
$7500-$9999	2.3	7.7
$10,000-$12,499	2.9	10.0
$12,500-$14,999	2.7	9.0
$15,000-$19,999	4.2	14.2
$20,000-$24,999	5.3	13.7
$25,000-$29,999	5.3	11.1
$30,000-$34,999	5.7	6.4
$35,000-$39,999	5.4	5.4
$40,000-$49,999	8.1	4.8
$50,000 or more	53.6	2.5
Employment status, %		
Employed	61.5	42.2
Unemployed	2.9	5.1
Not in Labor force	35.6	52.7
Household size, mean	2.48	2.70

Table 3. Estimated prevalence of households experiencing food security pre- and during the COVID-19 pandemic overall and for households with incomes less that approximately 185% of the Federal Poverty Threshold estimated from pooled data from the Current Population Survey Food Security Supplement from 2015 to 2021

Rural Classification*	Prevalence of Households Experiencing Food Security		Adjusted[1] Prevalence of Households Experiencing Food Security	
	2015-2019	2020-2021	2015-2019	2020-2021
All households				
Urban	87.7 [87.5-88.0]	88.8 [88.4-89.3]	87.5 [87.2-87.7]	88.6 [88.2-89.1]
Large Rural Town/City	85.5 [84.7-86.2]	87.1 [85.7-88.3]	86.7 [86.0-87.5]	88.3 [87.1-89.4]
Small Rural Town	82.8 [81.5-84.1]	87.3 [85.7-89.2]	85.3 [84.3-86.4]	89.1 [87.4-90.8]
Isolated Rural Town/Area	87.6 [86.3-88.8]	90.9 [88.7-92.7]	87.9 [86.8-89.1]	90.7 [88.8-92.6]
Low-income households				
Urban	72.7 [72.0-73.4]	74.0 [72.7-75.2]	72.9 [72.2-73.5]	74.1 [72.9-75.3]
Large Rural Town/City	72.2 [70.6-73.8]	74.9 [71.9-77.6]	71.5 [69.8-73.2]	74.6 [71.7-77.5]
Small Rural Town	70.2 [67.8-72.7]	76.9 [72.6-80.8]	69.8 [67.4-72.2]	76.8 [72.7-80.9]
Isolated Rural Town/Area	75.2 [72.5-77.7]	81.2 [76.1-85.3]	73.6 [70.8-76.3]	79.1 [74.3-84.0]

*Rural classification is based on secondary United States Department of Agriculture Rural Urban Commuting Area codes, which consider urbanicity, population size, and primary and secondary daily commuting flows

1.Adjusted prevalences were obtained from marginal effects based on regression models predicting food security status as a function of rural status, time period, their interaction, and adjusting for state of residence, household size, age of household respondents, sex of household respondent, racial self-classification of household respondent, Hispanic ethnicity of household respondent, household income, and employment status of household respondents.

Appendix 1

Table A1. Regression results from model predicting food security status as a function of rural status based on urbanicity, population size, and primary and secondary daily commuting flows, time period, and their interaction and adjusting for state of residence and demographic characteristics among all households

Variable	Odds Ratio	STD ERR	95% CI LB	95% CI UB
Rural Classification (Ref=Urban)				
Large Rural Town/City	0.93	0.04	0.86	1.00
Small Rural Town	0.81	0.04	0.73	0.89
Isolated Rural Town/Area	1.05	0.07	0.92	1.19
Time Period (Ref=2015-2019)				
2020-2021	1.00	0.03	0.94	1.06
Rural Classification x Time Period				
Large Rural Town x 2020-2021	1.04	0.08	0.89	1.21
Small Rural Town x 2020-2021	1.31	0.15	1.05	1.65
Isolated Rural Town/Area x 2020-2021	1.24	0.18	0.93	1.64
Age (Ref=34 and under)				
35 to 59	0.82	0.02	0.77	0.86
60 and older	1.88	0.07	1.75	2.02
Sex (Ref=Male)				
Female	0.76	0.02	0.72	0.79
Racial self-classification (Ref=white)				
Black	0.53	0.02	0.49	0.56
Asian, American Indian/Alaska Native, Native Hawaiian/Pacific Islander	1.12	0.06	1.00	1.25
Other	0.77	0.02	0.73	0.82
Hispanic ethnicity (Ref=non-Hispanic)				
Hispanic	0.72	0.02	0.68	0.77
Income category (Ref=less than $5,000)				
$5000-$7499	0.81	0.06	0.70	0.94
$7500-$9999	0.62	0.04	0.55	0.71
$10,000-$12,499	0.79	0.05	0.70	0.90
$12,500-$14,999	0.86	0.06	0.76	0.98
$15,000-$19,999	0.98	0.06	0.87	1.10
$20,000-$24,999	1.22	0.07	1.09	1.37
$25,000-$29,999	1.50	0.09	1.33	1.68
$30,000-$34,999	1.76	0.11	1.57	1.98
$35,000-$39,999	2.06	0.13	1.82	2.33
$40,000-$49,999	2.55	0.15	2.27	2.86
$50,000 or more	7.79	0.42	7.01	8.67
Employment status (Ref=employed)				
Unemployed	0.48	0.03	0.43	0.53
Not in the labor force	0.82	00.2	0.77	0.86
Household size	0.88	0.1	0.87	0.90

Notes: Model also included fixed effects for state of residence

Table A2. Regression results from model predicting food security status as a function of rural status based on urbanicity, population size, and primary and secondary daily commuting flows, time period, and their interaction and adjusting for state of residence and demographic characteristics among households incomes approximately less than 185% of the Federal Poverty Threshold

Variable	Odds Ratio	STD ERR	95% CI LB	95% CI UB
Rural Classification (Ref=Urban)				
Large Rural Town/City	0.93	0.05	0.85	1.03
Small Rural Town	0.85	0.06	0.75	0.97
Isolated Rural Town/Area	1.04	0.09	0.89	1.21
Time Period (Ref=2015-2019)				
2020-2021	1.01	0.04	0.94	1.09
Rural Classification x Time Period				
Large Rural Town x 2020-2021	1.10	0.11	0.91	1.34
Small Rural Town x 2020-2021	1.36	0.19	1.03	1.80
Isolated Rural Town/Area x 2020-2021	1.30	0.23	0.92	1.83
Age (Ref=34 and under)				
35 to 59	0.70	0.03	0.65	0.75
60 and older	1.59	0.07	1.46	1.74
Sex (Ref=Male)				
Female	0.79	0.02	0.75	0.84
Racial self-classification (Ref=white)				
Black	0.62	0.03	0.57	0.67
Asian, American Indian/Alaska Native, Native Hawaiian/Pacific Islander	1.16	0.08	1.01	1.33
Other	0.77	0.03	0.71	0.83
Hispanic ethnicity (Ref=non-Hispanic)				
Hispanic	0.88	0.04	0.82	0.96
Income category (Ref=less than $5,000)				
$5000-$7499	0.82	0.06	0.71	0.95
$7500-$9999	0.66	0.04	0.58	0.75
$10,000-$12,499	0.84	0.05	0.74	0.95
$12,500-$14,999	0.92	0.06	0.81	1.04
$15,000-$19,999	1.04	0.06	0.93	1.17
$20,000-$24,999	1.24	0.08	1.10	1.39
$25,000-$29,999	1.43	0.09	1.26	1.62
$30,000-$34,999	1.40	0.10	1.21	1.62
$35,000-$39,999	1.71	0.14	1.45	2.01
$40,000-$49,999	1.81	0.16	1.52	2.15
$50,000 or more	2.16	0.26	1.71	2.72
Employment status (Ref=employed)				
Unemployed	0.56	0.04	0.51	0.66
Not in the labor force	0.89	0.03	0.83	0.95
Household size	0.94	0.01	0.92	0.96

Notes: Model also included fixed effects for state of residence